Afsaneh Shams

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Objective:

Machine learning researcher with expertise in lightweight neural networks, generative AI, and large-scale data processing, particularly in healthcare domains. Seeking a research scientist or machine learning engineer role in industry to design scalable and interpretable AI solutions. Dedicated to bridging academic innovation with practical deployment to enhance decision-making, operational efficiency, and societal impact.

Education:

Ph.D. in Computer Science **Aug. 2020-Present**
University of Georgia, Athens, USA, GPA: 3.77

M.Sc. from Electrical and Computer Engineering Department **Sep. 2012-March 2016**
Shiraz University, Shiraz

B.Sc. from Electrical and Computer Engineering Department **Sep. 2007-Sep. 2011**
Shiraz University, Shiraz

Technical Interests:

- Machine Learning and Deep Learning
- Data Science and Statistical Analysis
- Large-Scale Data Processing and Real-World Data Analytics
- Evolutionary Algorithms and Optimization Techniques
- Generative AI and Large Language Models (LLMs)
- Hybrid Neural Architectures
- Computer Vision and Image Classification
- AI Applications in Healthcare and Education
- High-Performance Computing (HPC) and Model Evaluation under Class Imbalance

Work Experience:

Research Assistant, *Public Health Department, University of Georgia, Athens, GA, USA* **Aug. 2024-Present**

- Conduct large-scale analysis of CMS datasets (MedPAR, MDS, MBSF) to investigate hospital readmissions following skilled nursing facility (SNF) discharge.
- Develop scalable pipelines in Python for data cleaning, integration, and custom feature engineering.
- Apply advanced resampling techniques (e.g., SMOTE, SMOTEENN) to address extreme class imbalance.
- Build and evaluate predictive models, including Random Forest, XGBoost, Decision Trees, Linear Regression, and deep neural networks.
- Utilize high-performance computing clusters for parallel processing of millions of records.
- Deliver encrypted and publication-ready outputs for collaborative research projects.
- Specialize in deploying real-world machine learning solutions for structured healthcare data.

Research and Development Scientist-Intern, *Roocreations Company, San Jose, CA, USA* **June. 2024-Sep. 2024**

- Collaborated directly with the CEO and a faculty member from the University of San Francisco on AI model design and experimentation.
- Co-developed a novel gradient-descent-based neural network architecture using Python.
- Contributed to model optimization, debugging, and parameter flow design.
- Automated data handling and reporting using NumPy, Pandas, and internal file integration tools.
- Gained hands-on experience in AI innovation and early-stage neural architecture research in an industry setting.

Teaching Assistant, *University of Georgia, Athens, GA, USA* **Aug. 2020-Jan. 2025**

- Served as TA for multiple undergraduate courses including Numerical Simulations in Science & Engineering, Foundations for Informatics and Data Analytics, and Discrete Mathematics for Computer Science
- Acted as lab instructor for several semesters in CSCI 1301L (Java lab), leading each session with a short lecture to explain objectives and key programming concepts.
- Assisted students in understanding lab material, debugging code, and completing assignments.
- Used Eclipse, IntelliJ, and eLC (UGA's learning platform) to manage coursework and student communication.
- Consistently received positive evaluations from instructors and students.
- Recipient of the Outstanding Graduate Teaching Assistant Award (May 2023).
- Gained deep experience in academic mentoring, student engagement, and technical instruction.

Design Electrical Engineer, *PIDEC, Shiraz* **June 2012-March 2018**

- Contributed to 10–15 engineering projects across industrial systems.
- Designed and implemented electrical infrastructure for Heat Tracing Systems, Lighting Systems, Communication Systems, and CCTV and security networks.

Projects:

- **Healthcare Data Modeling:** Developed regression models on CMS Chronic Conditions Warehouse (CCW) data to analyze healthcare patterns and predict patient outcomes. Tasks included large-scale data cleaning, feature engineering, and predictive modeling.
- **KANEATFormer:** Designed and implemented KANEATFormer by integrating Kolmogorov–Arnold Networks (KAN, 2024) into EATFormer, an evolutionary transformer. Improved model interpretability and accuracy while reducing computational cost through learnable edge-based activations.
- **Hybrid Attention CNN (AECNNB):** Enhanced an evolutionary CNN (ECNNB) with CBAM and MobileViTv2 attention mechanisms, resulting in a 9.77% accuracy gain on CIFAR-10.
- **Evolutionary CNN:** Developed a highly accurate CNN optimized via evolutionary strategies, achieving 99.58% (EMNIST_Digits), 99.32% (MNIST), and 92.58% (Fashion-MNIST).
- **Neural Architecture Search:** Implemented an evolutionary algorithm to optimize the number of hidden layers and units for image classification tasks.
- **Deepfake Detection:** Used YOLOv3 to detect and annotate manipulated regions in PSCC-Net deepfake images with masks and bounding boxes.
- **COVID-GAN:** Applied both supervised and semi-supervised GANs to classify COVID-19 from chest X-rays using a mix of annotated and unannotated images.
- **RNA & EEG Classification:** Conducted analysis in WEKA for RNA structure prediction (94.01% accuracy) and EEG signal classification using Random Forest, Naïve Bayes, KNN, and Neural Networks.
- **Web-Based Tools:** Developed databases and backends for a project management tool (link) and a cinema ticketing system (link) using PostgreSQL, MySQL, and phpMyAdmin.
- **Collaborative Coding Platform:** Built a real-time synchronous programming environment to support cooperative learning and live code sharing (link).

Skills:

- Programming: Python, Java, Matlab
- Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, imbalanced-learn
- ML/DL Techniques: KAN (Kolmogorov–Arnold Network), Hybrid Architectures, Convolutional Neural Networks (CNN), Transformers, Regression, SMOTE, SMOTEENN, Hyperparameter Tuning
- Data Analysis: NumPy, Pandas, WEKA
- High-Performance Computing: SLURM, ProcessPoolExecutor, Pandarallel, Shell Scripting, HPC Cluster Usage
- Databases & Data Management: PostgreSQL, MySQL, phpMyAdmin, Neo4j
- Development Tools & Environments: Git, Linux, UNIX
- Modeling & Simulation Tools: Figma, StarUML, VHDL, Pspice, NS, SAFT
- Languages: English (Fluent), French (Basic)
- Core Competencies: Self-motivated, analytical, detail-oriented, collaborative, quick learner, effective communicator, strong multitasking

Publications:

- Shams, A., Bozorgi, E., Darbandi, M. R. (Nima), Amirian, S., & Rasheed, K. (2025). 'Evolutionary computing applications and trends from 2010 to 2025: A brief review'. Accepted at AIR-RES 2025.
- Bozorgi, E., Alqaaidi, S. K., Shams, A., et al. (2025). 'A survey on the recent random walk-based methods for embedding graphs. Journal of Supercomputing', 81, 619. https://doi.org/10.1007/s11227-025-07019-x
- Shams, A., Becker, K., Becker, D., Amirian, S., & Rasheed, K. (2024). 'Evolutionary CNN-based architectures with attention mechanisms for enhanced image classification'. In Artificial Intelligence: Machine Learning, Convolutional Neural Networks and Large Language Models, 1, 107.
- Shams, A., Becker, D., Becker, K., Amirian, S., & Rasheed, K. (2023). 'Evolving efficient CNN-based model for image classification. In 2023 Congress on Computer Science', Computer Engineering, & Applied Computing (CSCE), IEEE, pp. 228–235.
- Alqaaidi, S. K., Bozorgi, E., Shams, A., & Kochut, K. J. 'A few-shot learning-focused survey on recent named entity recognition and relation classification models'. (Manuscript).
- Shams, A., & Rasheed, K. 'Hybrid evolutionary KAN transformer'. (In preparation).
- Shams, A., & Rasheed, K. 'A survey of recent evolutionary algorithms'. (In preparation).

Professional Activities:

- Judge, Georgia Science & Engineering Fair (GSEF), March 2023
- Judge, Georgia Science & Engineering Fair (GSEF), April 2024
- Peer Reviewer, Neural Networks, 2025
- Peer Reviewer, Engineering Applications of Artificial Intelligence, 2025
- Peer Reviewer, Neurocomputing, 2025
- Conference Session Chair, The 2025 International Conference on the AI Revolution: Research, Ethics, and Society (AIR-RES 2025), USA, April 2025